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Raghavan et al.

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(54) **SELF-INTERFERENCE FEEDBACK FOR
MULTIPLE ANTENNA PANELS**

(71) Applicant: **QUALCOMM Incorporated**, San
Diego, CA (US)

(72) Inventors: **Vasanthan Raghavan**, West Windsor
Township, NJ (US); **Tianyang Bai**,
Somerville, NJ (US); **Jung Ho Ryu**,
Fort Lee, NJ (US)

(73) Assignee: **QUALCOMM Incorporated**, San
Diego, CA (US)

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H04L 5/14 (2006.01)

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CPC **H04B 17/345** (2015.01); **H04L 5/14**
(2013.01)

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H04L 5/14; H04L 5/0023; H04W 72/535
See application file for complete search history.

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Primary Examiner — Ayaz R Sheikh

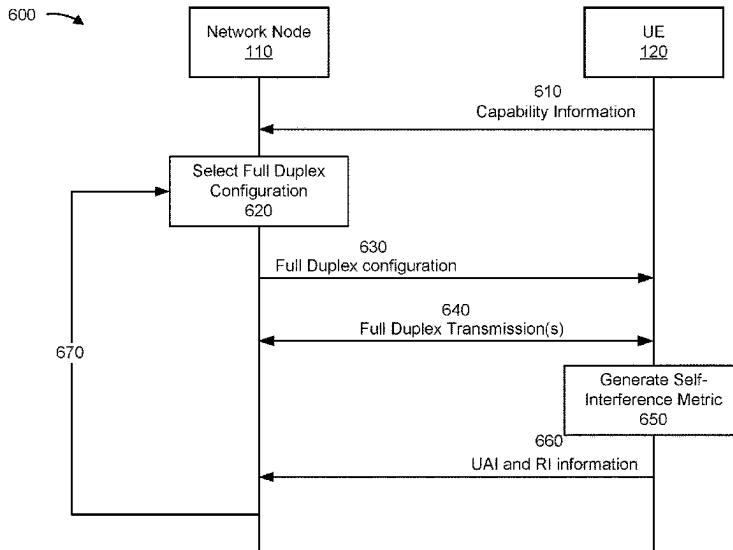
Assistant Examiner — Louis Samara

(74) *Attorney, Agent, or Firm* — Harrity & Harrity, LLP

(57) **ABSTRACT**

Various aspects of the present disclosure generally relate to
wireless communication. In some aspects, a user equipment
(UE) may receive a beam based at least in part on using at
least a first antenna panel. The UE may transmit a rank
indicator and UE assistance information (UAI), the UAI
comprising a self-interference metric associated with oper-
ating in a full duplex mode based at least in part on using the
at least first antenna panel. Numerous other aspects are
described.

30 Claims, 10 Drawing Sheets



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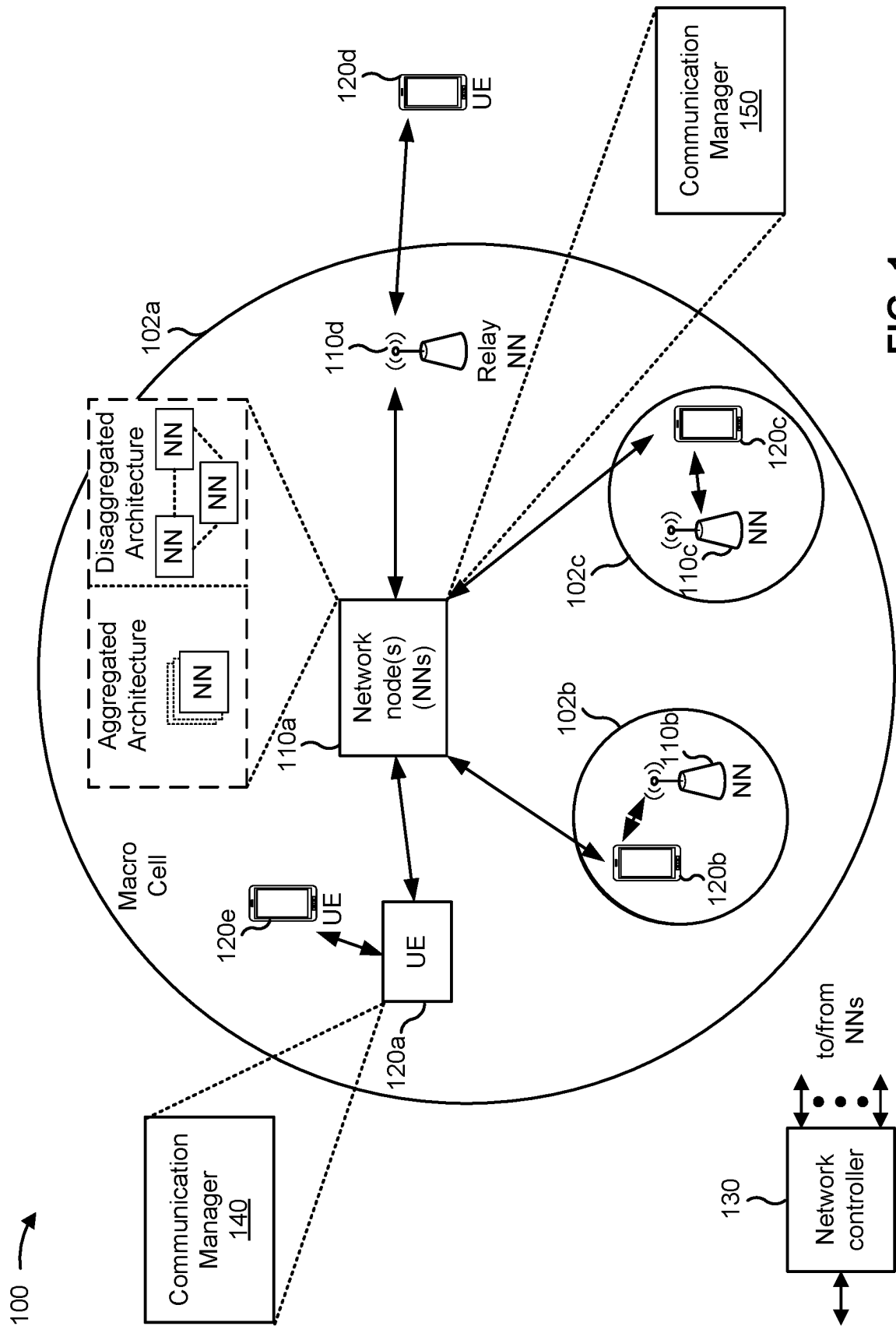


FIG. 1

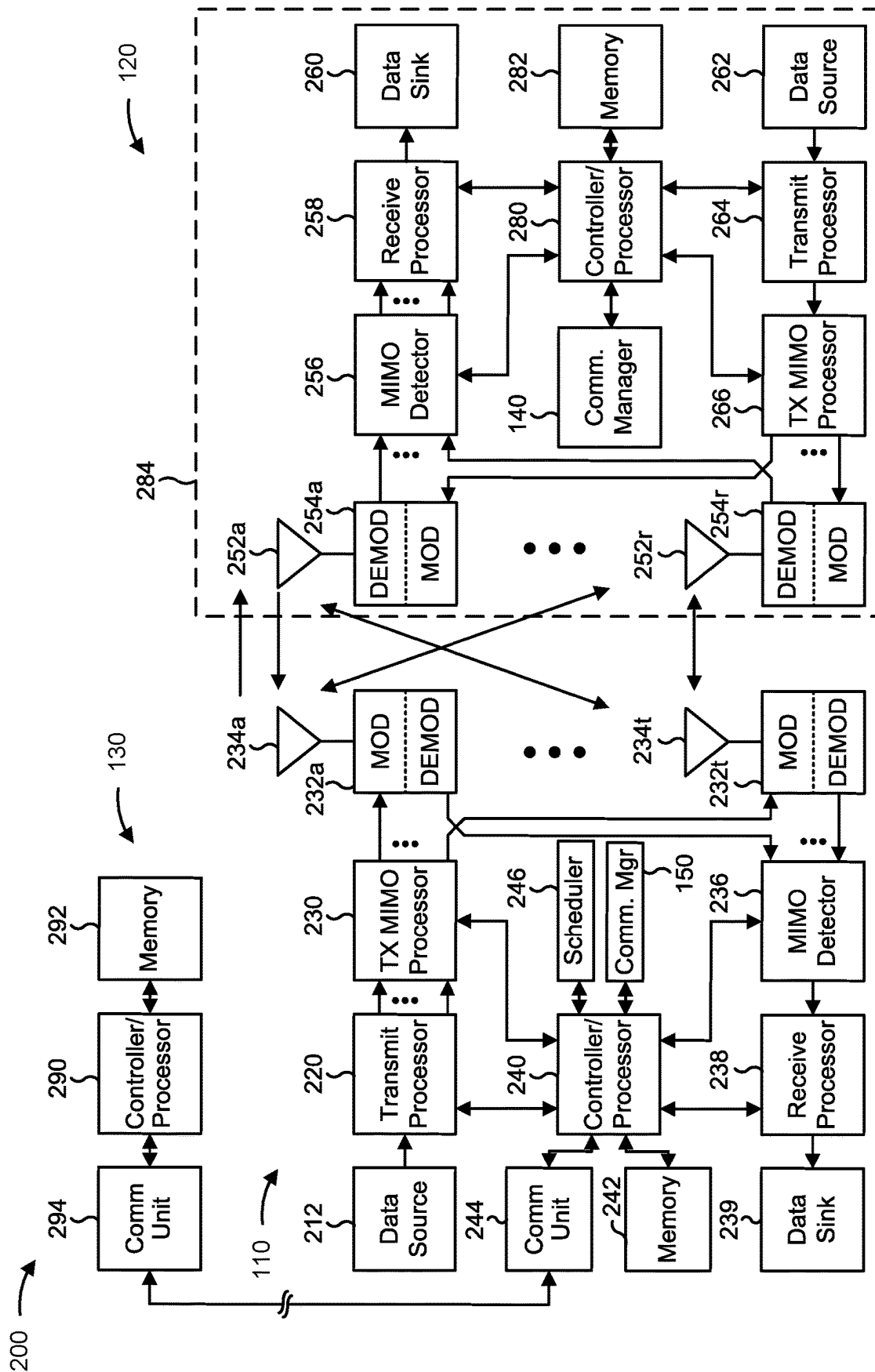


FIG. 2

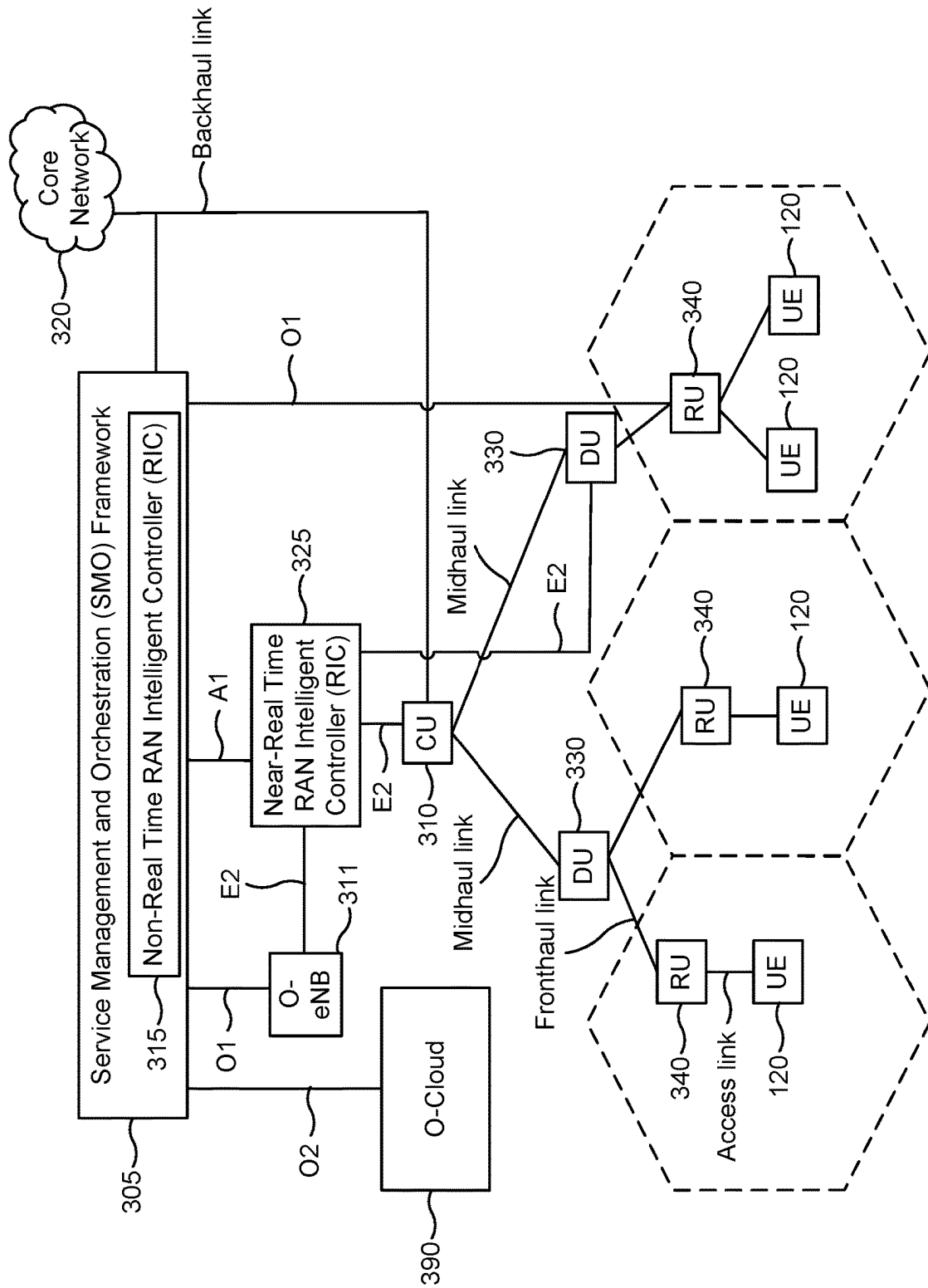


FIG. 3

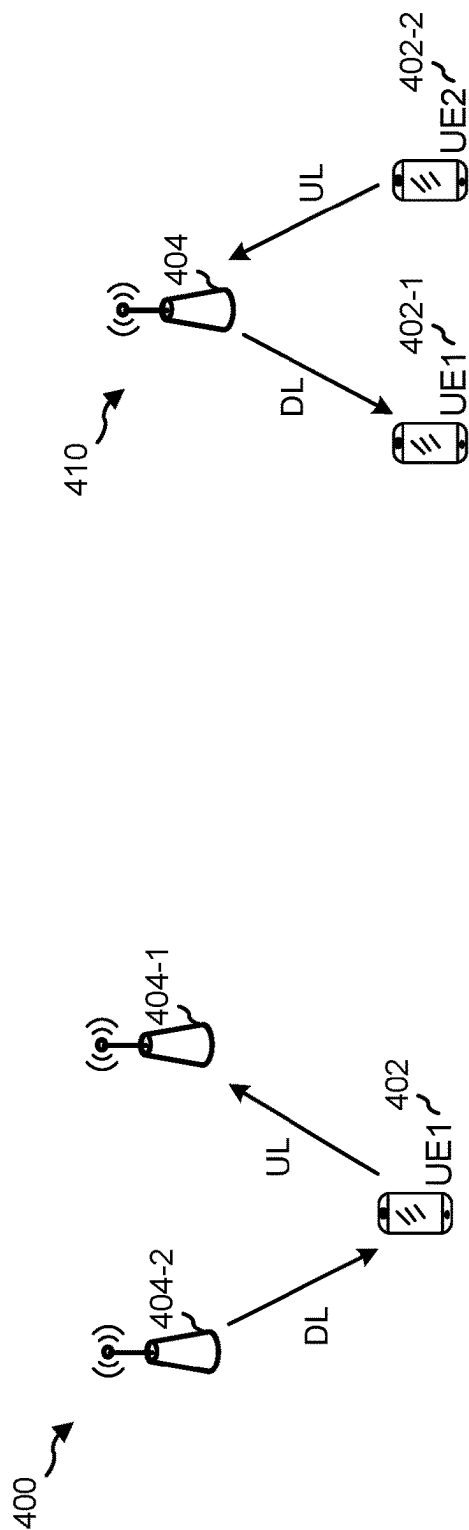


FIG. 4A

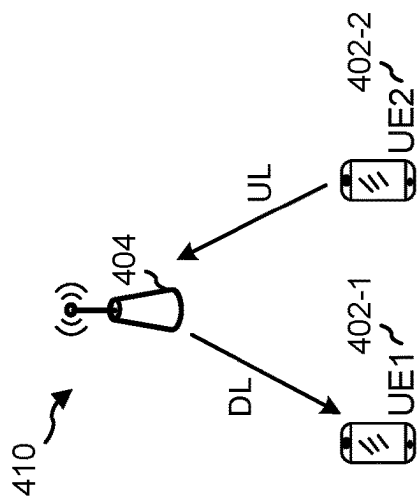


FIG. 4B

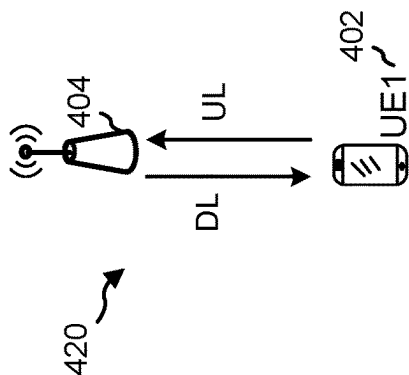


FIG. 4C

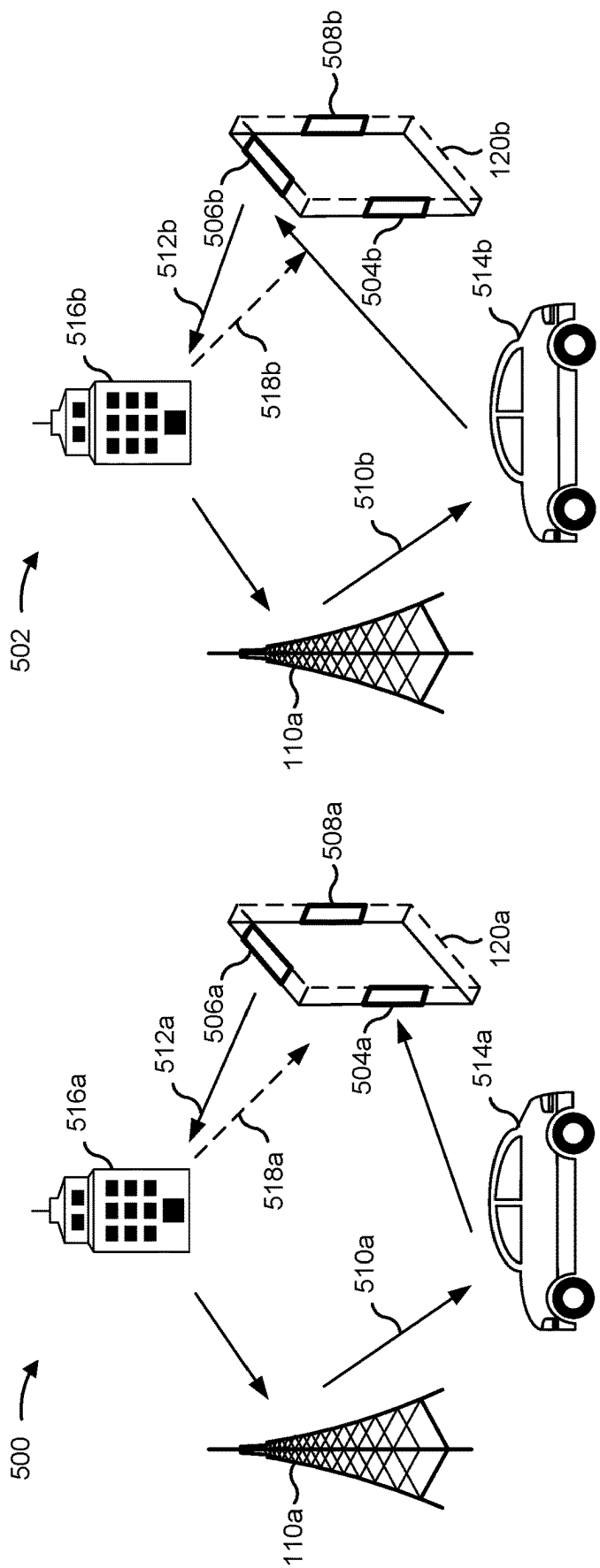


FIG. 5B

FIG. 5A

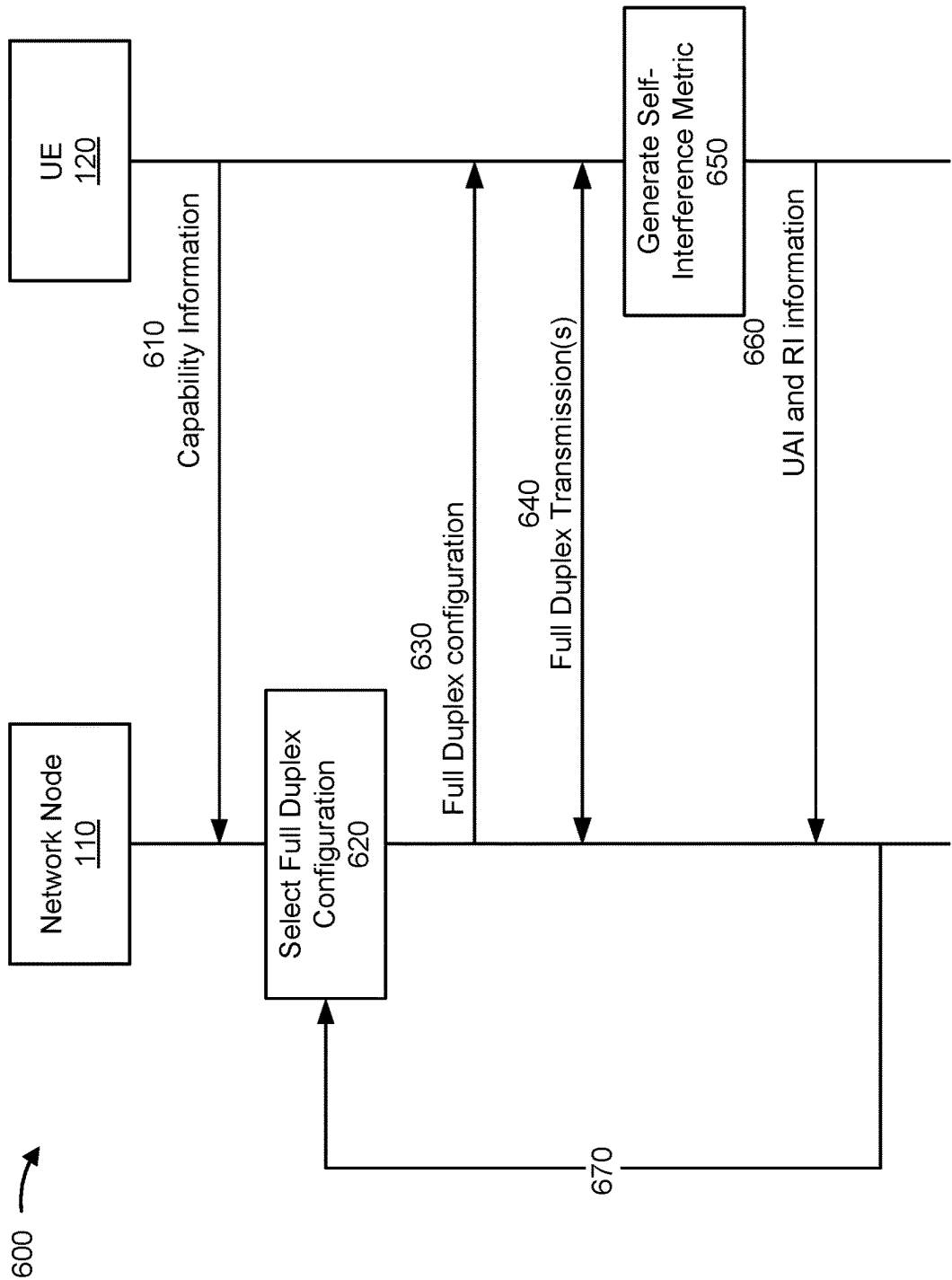


FIG. 6

700 →

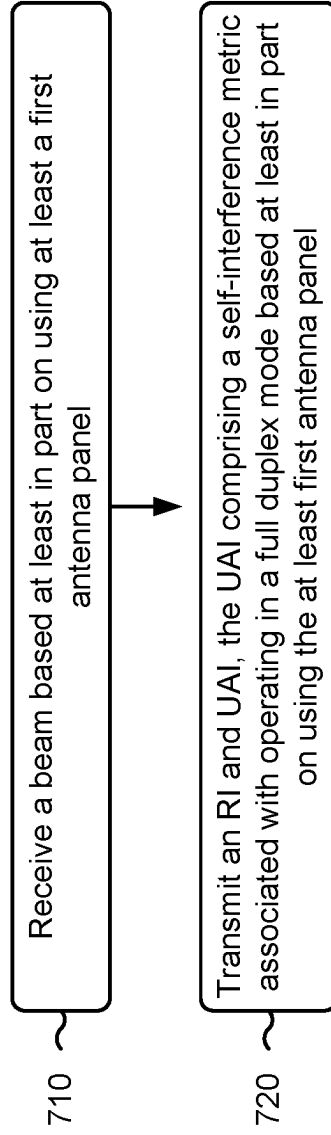


FIG. 7

800

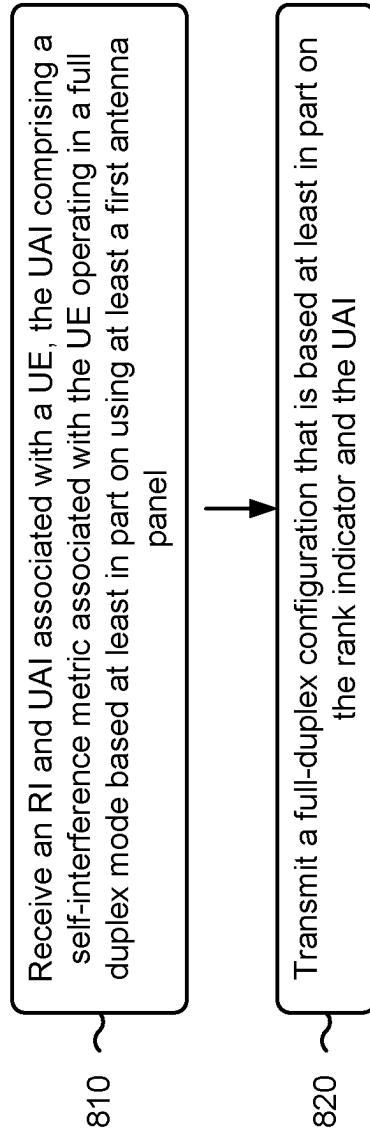


FIG. 8

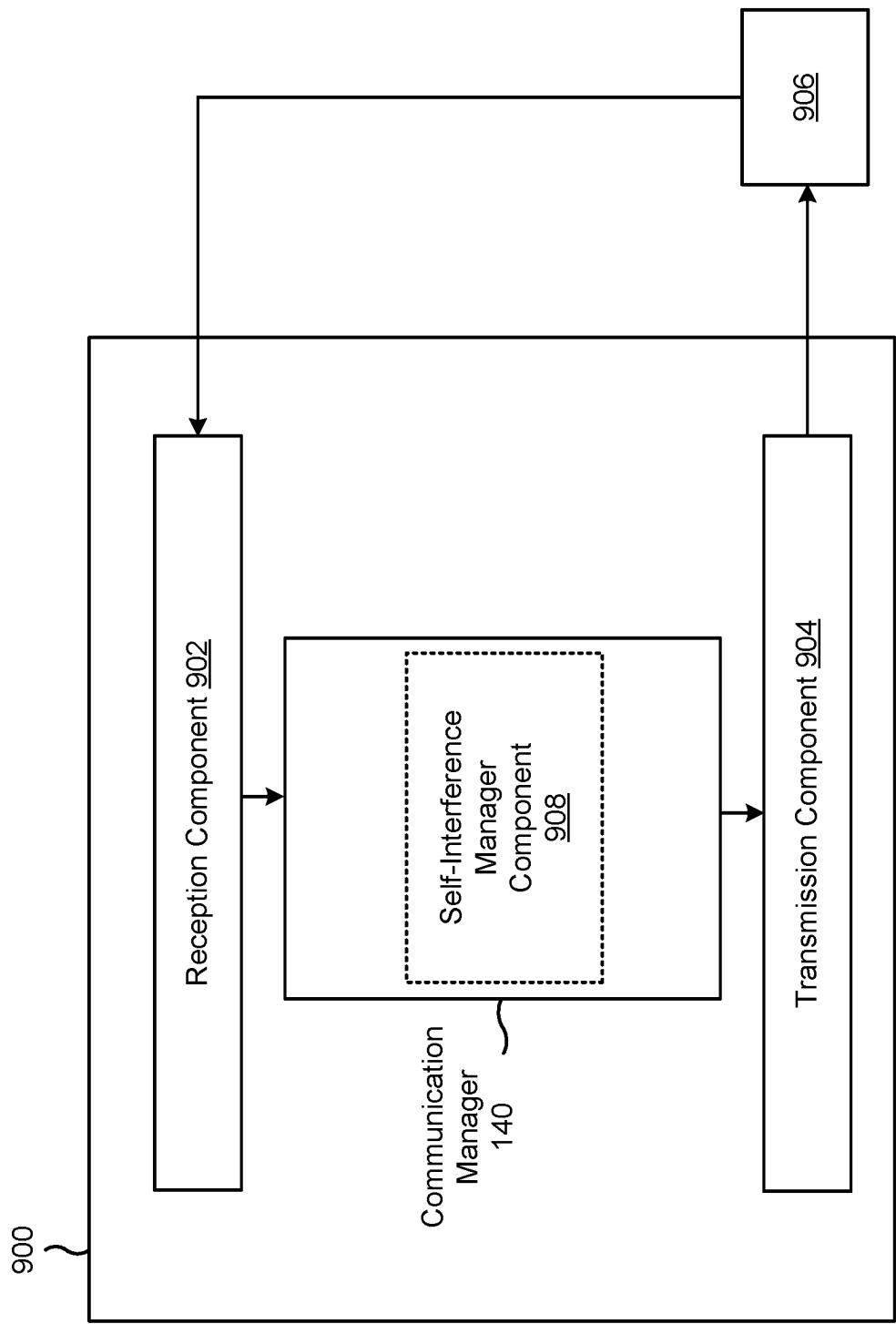


FIG. 9

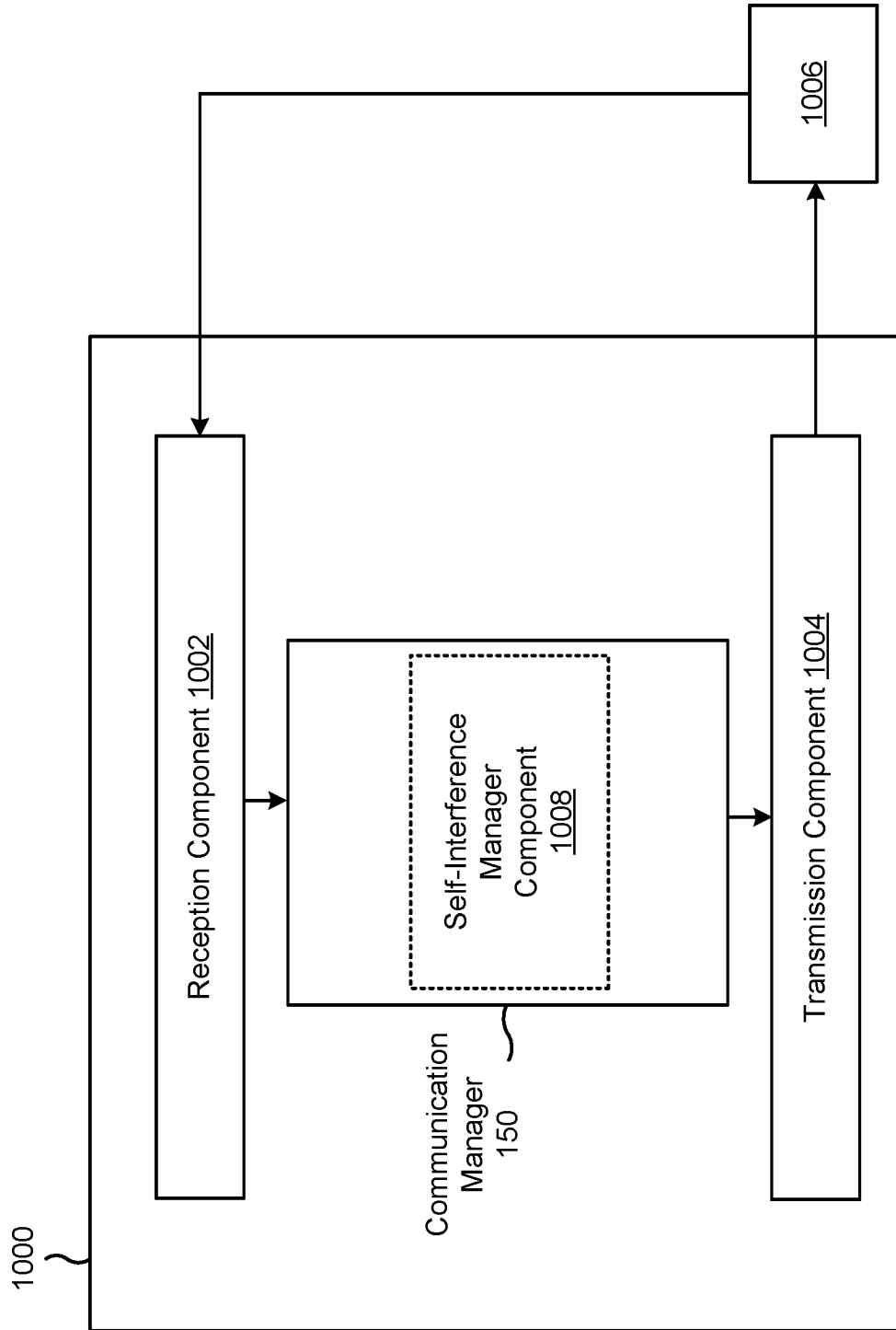


FIG. 10

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SELF-INTERFERENCE FEEDBACK FOR MULTIPLE ANTENNA PANELS

FIELD OF THE DISCLOSURE

Aspects of the present disclosure generally relate to wireless communication and to techniques and apparatuses for self-interference feedback for multiple antenna panels.

BACKGROUND

Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources (e.g., bandwidth, transmit power, or the like). Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, time division synchronous code division multiple access (TD-SCDMA) systems, and Long Term Evolution (LTE). LTE/LTE-Advanced is a set of enhancements to the Universal Mobile Telecommunications System (UMTS) mobile standard promulgated by the Third Generation Partnership Project (3GPP).

A wireless network may include one or more network nodes that support communication for wireless communication devices, such as a user equipment (UE) or multiple UEs. A UE may communicate with a network node via downlink communications and uplink communications. “Downlink” (or “DL”) refers to a communication link from the network node to the UE, and “uplink” (or “UL”) refers to a communication link from the UE to the network node. Some wireless networks may support device-to-device communication, such as via a local link (e.g., a sidelink (SL)), a wireless local area network (WLAN) link, and/or a wireless personal area network (WPAN) link, among other examples).

The above multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different UEs to communicate on a municipal, national, regional, and/or global level. New Radio (NR), which may be referred to as 5G, is a set of enhancements to the LTE mobile standard promulgated by the 3GPP. NR is designed to better support mobile broadband internet access by improving spectral efficiency, lowering costs, improving services, making use of new spectrum, and better integrating with other open standards using orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) (CP-OFDM) on the downlink, using CP-OFDM and/or single-carrier frequency division multiplexing (SC-FDM) (also known as discrete Fourier transform spread OFDM (DFT-s-OFDM)) on the uplink, as well as supporting beamforming, multiple-input multiple-output (MIMO) antenna technology, and carrier aggregation. As the demand for mobile broadband access continues to increase, further improvements in LTE, NR, and other radio access technologies remain useful.

SUMMARY

Some aspects described herein relate to a method of wireless communication performed by a user equipment

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(UE). The method may include receiving a beam based at least in part on using at least a first antenna panel. The method may include transmitting a rank indicator (RI) and UE assistance information (UAI), the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel.

Some aspects described herein relate to a method of wireless communication performed by a network node. The method may include receiving an RI and UAI associated with a UE, the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel. The method may include transmitting full duplex configuration information that is based at least in part on the rank indicator and the UAI.

Some aspects described herein relate to an apparatus for wireless communication at a UE. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to cause the apparatus to receive a beam based at least in part on using at least a first antenna panel. The one or more processors may be configured to cause the apparatus to transmit an RI and UAI, the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel.

Some aspects described herein relate to an apparatus for wireless communication at a network node. The apparatus may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to cause the apparatus to receive an RI and UAI associated with a UE, the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel. The one or more processors may be configured to transmit full duplex configuration information that is based at least in part on the rank indicator and the UAI.

Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a UE. The set of instructions, when executed by one or more processors of the UE, may cause the UE to receive a beam based at least in part on using at least a first antenna panel. The set of instructions, when executed by one or more processors of the UE, may cause the UE to transmit an RI and UAI, the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel.

Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a network node. The set of instructions, when executed by one or more processors of the network node, may cause the network node to receive an RI and UAI associated with a UE, the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel. The set of instructions, when executed by one or more processors of the network node, may cause the network node to transmit full duplex configuration information that is based at least in part on the rank indicator and the UAI.

Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving a beam based at least in part on using at least a first antenna panel. The apparatus may include means for transmitting an RI and UAI, the UAI comprising a self-

interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel.

Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving an RI and UAI associated with a UE, the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel. The apparatus may include means for transmitting full duplex configuration information that is based at least in part on the rank indicator and the UAI.

Aspects generally include a method, apparatus, system, computer program product, non-transitory computer-readable medium, user equipment, base station, network entity, network node, wireless communication device, and/or processing system as substantially described herein with reference to and as illustrated by the drawings and specification.

The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter. The conception and specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages, will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purposes of illustration and description, and not as a definition of the limits of the claims.

While aspects are described in the present disclosure by illustration to some examples, those skilled in the art will understand that such aspects may be implemented in many different arrangements and scenarios. Techniques described herein may be implemented using different platform types, devices, systems, shapes, sizes, and/or packaging arrangements. For example, some aspects may be implemented via integrated chip embodiments or other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, and/or artificial intelligence devices). Aspects may be implemented in chip-level components, modular components, non-modular components, non-chip-level components, device-level components, and/or system-level components. Devices incorporating described aspects and features may include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals may include one or more components for analog and digital purposes (e.g., hardware components including antennas, radio frequency (RF) chains, power amplifiers, modulators, buffers, processors, interleavers, adders, and/or summers). It is intended that aspects described herein may be practiced in a wide variety of devices, components, systems, distributed arrangements, and/or end-user devices of varying size, shape, and constitution.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to

aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects. The same reference numbers in different drawings may identify the same or similar elements.

FIG. 1 is a diagram illustrating an example of a wireless network, in accordance with the present disclosure.

FIG. 2 is a diagram illustrating an example of a network node in communication with a user equipment (UE) in a wireless network, in accordance with the present disclosure.

FIG. 3 is a diagram illustrating an example disaggregated base station architecture, in accordance with the present disclosure.

FIGS. 4A-4C are diagrams illustrating examples of full duplex communication in accordance with the present disclosure.

FIGS. 5A and 5B are diagrams illustrating a first example and a second example of full duplex communications based at least in part on multiple antenna panels, in accordance with the present disclosure.

FIG. 6 is a diagram illustrating an example of a wireless communication process between a network node and a UE, in accordance with the present disclosure.

FIG. 7 is a diagram illustrating an example process performed, for example, by a UE, in accordance with the present disclosure.

FIG. 8 is a diagram illustrating an example process performed, for example, by a network node, in accordance with the present disclosure.

FIG. 9 is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.

FIG. 10 is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.

DETAILED DESCRIPTION

Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. One skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

Several aspects of telecommunication systems will now be presented with reference to various apparatuses and techniques. These apparatuses and techniques will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes, algorithms, or the like (collectively referred to as “elements”). These elements may be implemented using hardware, software, or

combinations thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

While aspects may be described herein using terminology commonly associated with a 5G or New Radio (NR) radio access technology (RAT), aspects of the present disclosure can be applied to other RATs, such as a 3G RAT, a 4G RAT, and/or a RAT subsequent to 5G (e.g., 6G).

FIG. 1 is a diagram illustrating an example of a wireless network 100, in accordance with the present disclosure. The wireless network 100 may be or may include elements of a 5G (e.g., NR) network and/or a 4G (e.g., Long Term Evolution (LTE)) network, among other examples. The wireless network 100 may include one or more network nodes 110 (shown as a network node 110a, a network node 110b, a network node 110c, and a network node 110d), a user equipment (UE) 120 or multiple UEs 120 (shown as a UE 120a, a UE 120b, a UE 120c, a UE 120d, and a UE 120e), and/or other entities. A network node 110 is a network node that communicates with UEs 120. As shown, a network node 110 may include one or more network nodes. For example, a network node 110 may be an aggregated network node, meaning that the aggregated network node is configured to utilize a radio protocol stack that is physically or logically integrated within a single radio access network (RAN) node (e.g., within a single device or unit). As another example, a network node 110 may be a disaggregated network node (sometimes referred to as a disaggregated base station), meaning that the network node 110 is configured to utilize a protocol stack that is physically or logically distributed among two or more nodes (such as one or more central units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)).

In some examples, a network node 110 is or includes a network node that communicates with UEs 120 via a radio access link, such as an RU. In some examples, a network node 110 is or includes a network node that communicates with other network nodes 110 via a fronthaul link or a midhaul link, such as a DU. In some examples, a network node 110 is or includes a network node that communicates with other network nodes 110 via a midhaul link or a core network via a backhaul link, such as a CU. In some examples, a network node 110 (such as an aggregated network node 110 or a disaggregated network node 110) may include multiple network nodes, such as one or more RUs, one or more CUs, and/or one or more DUs. A network node 110 may include, for example, an NR base station, an LTE base station, a Node B, an eNB (e.g., in 4G), a gNB (e.g., in 5G), an access point, a transmission reception point (TRP), a DU, an RU, a CU, a mobility element of a network, a core network node, a network element, a network equipment, a RAN node, or a combination thereof. In some examples, the network nodes 110 may be interconnected to one another or to one or more other network nodes 110 in the wireless network 100 through various types of fronthaul, midhaul, and/or backhaul interfaces, such as a direct physical connection, an air interface, or a virtual network, using any suitable transport network.

In some examples, a network node 110 may provide communication coverage for a particular geographic area. In the Third Generation Partnership Project (3GPP), the term “cell” can refer to a coverage area of a network node 110 and/or a network node subsystem serving this coverage area, depending on the context in which the term is used. A network node 110 may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or another type of

cell. A macro cell may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs 120 with service subscriptions. A pico cell may cover a relatively small geographic area and may allow unrestricted access by UEs 120 with service subscriptions. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs 120 having association with the femto cell (e.g., UEs 120 in a closed subscriber group (CSG)). A network node 110 for a macro cell may be referred to as a macro network node. A network node 110 for a pico cell may be referred to as a pico network node. A network node 110 for a femto cell may be referred to as a femto network node or an in-home network node. In the example shown in FIG. 1, the network node 110a may be a macro network node for a macro cell 102a, the network node 110b may be a pico network node for a pico cell 102b, and the network node 110c may be a femto network node for a femto cell 102c. A network node may support one or multiple (e.g., three) cells. In some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a network node 110 that is mobile (e.g., a mobile network node).

In some aspects, the term “base station” or “network node” may refer to an aggregated base station, a disaggregated base station, an integrated access and backhaul (IAB) node, a relay node, or one or more components thereof. For example, in some aspects, “base station” or “network node” may refer to a CU, a DU, an RU, a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, or a combination thereof. In some aspects, the term “base station” or “network node” may refer to one device configured to perform one or more functions, such as those described herein in connection with the network node 110. In some aspects, the term “base station” or “network node” may refer to a plurality of devices configured to perform the one or more functions. For example, in some distributed systems, each of a quantity of different devices (which may be located in the same geographic location or in different geographic locations) may be configured to perform at least a portion of a function, or to duplicate performance of at least a portion of the function, and the term “base station” or “network node” may refer to any one or more of those different devices. In some aspects, the term “base station” or “network node” may refer to one or more virtual base stations or one or more virtual base station functions. For example, in some aspects, two or more base station functions may be instantiated on a single device. In some aspects, the term “base station” or “network node” may refer to one of the base station functions and not another. In this way, a single device may include more than one base station.

The wireless network 100 may include one or more relay stations. A relay station is a network node that can receive a transmission of data from an upstream node (e.g., a network node 110 or a UE 120) and send a transmission of the data to a downstream node (e.g., a UE 120 or a network node 110). A relay station may be a UE 120 that can relay transmissions for other UEs 120. In the example shown in FIG. 1, the network node 110d (e.g., a relay network node) may communicate with the network node 110a (e.g., a macro network node) and the UE 120d in order to facilitate communication between the network node 110a and the UE 120d. A network node 110 that relays communications may be referred to as a relay station, a relay base station, a relay network node, a relay node, a relay, or the like.

The wireless network **100** may be a heterogeneous network that includes network nodes **110** of different types, such as macro network nodes, pico network nodes, femto network nodes, relay network nodes, or the like. These different types of network nodes **110** may have different transmit power levels, different coverage areas, and/or different impacts on interference in the wireless network **100**. For example, macro network nodes may have a high transmit power level (e.g., 5 to 40 watts) whereas pico network nodes, femto network nodes, and relay network nodes may have lower transmit power levels (e.g., 0.1 to 2 watts).

A network controller **130** may couple to or communicate with a set of network nodes **110** and may provide coordination and control for these network nodes **110**. The network controller **130** may communicate with the network nodes **110** via a backhaul communication link or a midhaul communication link. The network nodes **110** may communicate with one another directly or indirectly via a wireless or wireline backhaul communication link. In some aspects, the network controller **130** may be a CU or a core network device, or may include a CU or a core network device.

The UEs **120** may be dispersed throughout the wireless network **100**, and each UE **120** may be stationary or mobile. A UE **120** may include, for example, an access terminal, a terminal, a mobile station, and/or a subscriber unit. A UE **120** may be a cellular phone (e.g., a smart phone), a personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a laptop computer, a cordless phone, a wireless local loop (WLL) station, a tablet, a camera, a gaming device, a netbook, a smartbook, an ultrabook, a medical device, a biometric device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, a smart wristband, smart jewelry (e.g., a smart ring or a smart bracelet)), an entertainment device (e.g., a music device, a video device, and/or a satellite radio), a vehicular component or sensor, a smart meter/sensor, industrial manufacturing equipment, a global positioning system device, a UE function of a network node, and/or any other suitable device that is configured to communicate via a wireless or wired medium.

Some UEs **120** may be considered machine-type communication (MTC) or evolved or enhanced machine-type communication (eMTC) UEs. An MTC UE and/or an eMTC UE may include, for example, a robot, a drone, a remote device, a sensor, a meter, a monitor, and/or a location tag, that may communicate with a network node, another device (e.g., a remote device), or some other entity. Some UEs **120** may be considered Internet-of-Things (IoT) devices, and/or may be implemented as NB-IoT (narrowband IoT) devices. Some UEs **120** may be considered a Customer Premises Equipment. A UE **120** may be included inside a housing that houses components of the UE **120**, such as processor components and/or memory components. In some examples, the processor components and the memory components may be coupled together. For example, the processor components (e.g., one or more processors) and the memory components (e.g., a memory) may be operatively coupled, communicatively coupled, electronically coupled, and/or electrically coupled.

In general, any number of wireless networks **100** may be deployed in a given geographic area. Each wireless network **100** may support a particular RAT and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic

area in order to avoid interference between wireless networks of different RATs. In some cases, NR or 5G RAT networks may be deployed.

In some examples, two or more UEs **120** (e.g., shown as UE **120a** and UE **120e**) may communicate directly using one or more sidelink channels (e.g., without using a network node **110** as an intermediary to communicate with one another). For example, the UEs **120** may communicate using peer-to-peer (P2P) communications, device-to-device (D2D) communications, a vehicle-to-everything (V2X) protocol (e.g., which may include a vehicle-to-vehicle (V2V) protocol, a vehicle-to-infrastructure (V2I) protocol, or a vehicle-to-pedestrian (V2P) protocol), and/or a mesh network. In such examples, a UE **120** may perform scheduling operations, resource selection operations, and/or other operations described elsewhere herein as being performed by the network node **110**.

Devices of the wireless network **100** may communicate using the electromagnetic spectrum, which may be subdivided by frequency or wavelength into various classes, bands, channels, or the like. For example, devices of the wireless network **100** may communicate using one or more operating bands. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

With the above examples in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like, if used herein, may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like, if used herein, may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band. It is contemplated that the frequencies included in these operating bands (e.g., FR1, FR2, FR3, FR4, FR4-a, FR4-1, and/or FR5) may be modified, and techniques described herein are applicable to those modified frequency ranges.

In some aspects, the UE **120** may include a communication manager **140**. As described in more detail elsewhere herein, the communication manager **140** may receive a beam based at least in part on using at least a first antenna panel;

and transmit a rank indicator (RI) and UE assistance information (UAI), the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel. Additionally, or alternatively, the communication manager **140** may perform one or more other operations described herein.

In some aspects, the network node may include a communication manager **150**. As described in more detail elsewhere herein, the communication manager **150** may receive an RI and UAI associated with a UE, the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel; and transmit full duplex configuration information that is based at least in part on the rank indicator and the UAI. Additionally, or alternatively, the communication manager **150** may perform one or more other operations described herein.

As indicated above, FIG. 1 is provided as an example. Other examples may differ from what is described with regard to FIG. 1.

FIG. 2 is a diagram illustrating an example **200** of a network node **110** in communication with a UE **120** in a wireless network **100**, in accordance with the present disclosure. The network node **110** may be equipped with a set of antennas **234a** through **234t**, such as T antennas ($T \geq 1$). The UE **120** may be equipped with a set of antennas **252a** through **252r**, such as R antennas ($R \geq 1$). The network node **110** of example **200** includes one or more radio frequency components, such as antennas **234** and a modem **254**. In some examples, a network node **110** may include an interface, a communication component, or another component that facilitates communication with the UE **120** or another network node. Some network nodes **110** may not include radio frequency components that facilitate direct communication with the UE **120**, such as one or more CUs, or one or more DUs.

At the network node **110**, a transmit processor **220** may receive data, from a data source **212**, intended for the UE **120** (or a set of UEs **120**). The transmit processor **220** may select one or more modulation and coding schemes (MCSs) for the UE **120** based at least in part on one or more channel quality indicators (CQIs) received from that UE **120**. The network node **110** may process (e.g., encode and modulate) the data for the UE **120** based at least in part on the MCS(s) selected for the UE **120** and may provide data symbols for the UE **120**. The transmit processor **220** may process system information (e.g., for semi-static resource partitioning information (SRPI)) and control information (e.g., CQI requests, grants, and/or upper layer signaling) and provide overhead symbols and control symbols. The transmit processor **220** may generate reference symbols for reference signals (e.g., a cell-specific reference signal (CRS) or a demodulation reference signal (DMRS)) and synchronization signals (e.g., a primary synchronization signal (PSS) or a secondary synchronization signal (SSS)). A transmit (TX) multiple-input multiple-output (MIMO) processor **230** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, the overhead symbols, and/or the reference symbols, if applicable, and may provide a set of output symbol streams (e.g., T output symbol streams) to a corresponding set of modems **232** (e.g., T modems), shown as modems **232a** through **232t**. For example, each output symbol stream may be provided to a modulator (MOD) component of a modem **232**. Each modem **232** may use a respective modulator component to process a respective output symbol stream (e.g., for OFDM) to obtain an output

sample stream. Each modem **232** may further use a respective modulator component to process (e.g., convert to analog, amplify, filter, and/or upconvert) the output sample stream to obtain a downlink signal. The modems **232a** through **232t** may transmit a set of downlink signals (e.g., T downlink signals) via a corresponding set of antennas **234** (e.g., T antennas), shown as antennas **234a** through **234t**.

At the UE **120**, a set of antennas **252** (shown as antennas **252a** through **252r**) may receive the downlink signals from the network node **110** and/or other network nodes **110** and may provide a set of received signals (e.g., R received signals) to a set of modems **254** (e.g., R modems), shown as modems **254a** through **254r**. For example, each received signal may be provided to a demodulator (DEMOD) component of a modem **254**. Each modem **254** may use a respective demodulator component to condition (e.g., filter, amplify, downconvert, and/or digitize) a received signal to obtain input samples. Each modem **254** may use a demodulator component to further process the input samples (e.g., for OFDM) to obtain received symbols. A MIMO detector **256** may obtain received symbols from the modems **254**, may perform MIMO detection on the received symbols if applicable, and may provide detected symbols. A receive processor **258** may process (e.g., demodulate and decode) the detected symbols, may provide decoded data for the UE **120** to a data sink **260**, and may provide decoded control information and system information to a controller/processor **280**. The term "controller/processor" may refer to one or more controllers, one or more processors, or a combination thereof. A channel processor may determine a reference signal received power (RSRP) parameter, a received signal strength indicator (RSSI) parameter, a reference signal received quality (RSRQ) parameter, and/or a CQI parameter, among other examples. In some examples, one or more components of the UE **120** may be included in a housing **284**.

The network controller **130** may include a communication unit **294**, a controller/processor **290**, and a memory **292**. The network controller **130** may include, for example, one or more devices in a core network. The network controller **130** may communicate with the network node **110** via the communication unit **294**.

One or more antennas (e.g., antennas **234a** through **234t** and/or antennas **252a** through **252r**) may include, or may be included within, one or more antenna panels, one or more antenna groups, one or more sets of antenna elements, and/or one or more antenna arrays, among other examples. An antenna panel, an antenna group, a set of antenna elements, and/or an antenna array may include one or more antenna elements (within a single housing or multiple housings), a set of coplanar antenna elements, a set of non-coplanar antenna elements, and/or one or more antenna elements coupled to one or more transmission and/or reception components, such as one or more components of FIG. 2.

On the uplink, at the UE **120**, a transmit processor **264** may receive and process data from a data source **262** and control information (e.g., for reports that include RSRP, RSSI, RSRQ, and/or CQI) from the controller/processor **280**. The transmit processor **264** may generate reference symbols for one or more reference signals. The symbols from the transmit processor **264** may be precoded by a TX MIMO processor **266** if applicable, further processed by the modems **254** (e.g., for DFT-s-OFDM or CP-OFDM), and transmitted to the network node **110**. In some examples, the modem **254** of the UE **120** may include a modulator and a demodulator. In some examples, the UE **120** includes a

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transceiver. The transceiver may include any combination of the antenna(s) 252, the modem(s) 254, the MIMO detector 256, the receive processor 258, the transmit processor 264, and/or the TX MIMO processor 266. The transceiver may be used by a processor (e.g., the controller/processor 280) and the memory 282 to perform aspects of any of the methods described herein (e.g., with reference to FIGS. 5-10).

At the network node 110, the uplink signals from UE 120 and/or other UEs may be received by the antennas 234, processed by the modem 232 (e.g., a demodulator component, shown as DEMOD, of the modem 232), detected by a MIMO detector 236 if applicable, and further processed by a receive processor 238 to obtain decoded data and control information sent by the UE 120. The receive processor 238 may provide the decoded data to a data sink 239 and provide the decoded control information to the controller/processor 240. The network node 110 may include a communication unit 244 and may communicate with the network controller 130 via the communication unit 244. The network node 110 may include a scheduler 246 to schedule one or more UEs 120 for downlink and/or uplink communications. In some examples, the modem 232 of the network node 110 may include a modulator and a demodulator. In some examples, the network node 110 includes a transceiver. The transceiver may include any combination of the antenna(s) 234, the modem(s) 232, the MIMO detector 236, the receive processor 238, the transmit processor 220, and/or the TX MIMO processor 230. The transceiver may be used by a processor (e.g., the controller/processor 240) and the memory 242 to perform aspects of any of the methods described herein (e.g., with reference to FIGS. 5-10).

The controller/processor 240 of the network node 110, the controller/processor 280 of the UE 120, and/or any other component(s) of FIG. 2 may perform one or more techniques associated with self-interference feedback for multiple antenna panels, as described in more detail elsewhere herein. For example, the controller/processor 240 of the network node 110, the controller/processor 280 of the UE 120, and/or any other component(s) of FIG. 2 may perform or direct operations of, for example, process 700 of FIG. 7, process 800 of FIG. 8, and/or other processes as described herein. The memory 242 and the memory 282 may store data and program codes for the network node 110 and the UE 120, respectively. In some examples, the memory 242 and/or the memory 282 may include a non-transitory computer-readable medium storing one or more instructions (e.g., code and/or program code) for wireless communication. For example, the one or more instructions, when executed (e.g., directly, or after compiling, converting, and/or interpreting) by one or more processors of the network node 110 and/or the UE 120, may cause the one or more processors, the UE 120, and/or the network node 110 to perform or direct operations of, for example, process 700 of FIG. 7, process 800 of FIG. 8, and/or other processes as described herein. In some examples, executing instructions may include running the instructions, converting the instructions, compiling the instructions, and/or interpreting the instructions, among other examples.

In some aspects, a UE (e.g., the UE 120) includes means for receiving a beam based at least in part on using at least a first antenna panel; and/or means for transmitting an RI and UAI, the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel. The means for the UE to perform operations described herein may include, for example, one or more of communication manager 140, antenna 252, modem 254, MIMO detector

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256, receive processor 258, transmit processor 264, TX MIMO processor 266, controller/processor 280, or memory 282.

In some aspects, a network node (e.g., the network node 110) includes means for receiving an RI and UAI associated with a UE, the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel; and/or means for transmitting full duplex configuration information that is based at least in part on the rank indicator and the UAI. In some aspects, the means for the network node to perform operations described herein may include, for example, one or more of communication manager 150, transmit processor 220, TX MIMO processor 230, modem 232, antenna 234, MIMO detector 236, receive processor 238, controller/processor 240, memory 242, or scheduler 246.

While blocks in FIG. 2 are illustrated as distinct components, the functions described above with respect to the blocks may be implemented in a single hardware, software, or combination component or in various combinations of components. For example, the functions described with respect to the transmit processor 264, the receive processor 258, and/or the TX MIMO processor 266 may be performed by or under the control of the controller/processor 280.

As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a RAN node, a core network node, a network element, a base station, or a network equipment may be implemented in an aggregated or disaggregated architecture. For example, a base station (such as a Node B (NB), an evolved NB (eNB), an NR base station (BS), a 5G NB, an access point (AP), a TRP, or a cell, among other examples), or one or more units (or one or more components) performing base station functionality, may be implemented as an aggregated base station (also known as a standalone base station or a monolithic base station) or a disaggregated base station. "Network entity" or "network node" may refer to a disaggregated base station, or to one or more units of a disaggregated base station (such as one or more CUs, one or more DUs, one or more RUs, or a combination thereof).

An aggregated base station (e.g., an aggregated network node) may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node (e.g., within a single device or unit). A disaggregated base station (e.g., a disaggregated network node) may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more CUs, one or more DUs, or one or more RUs). In some examples, a CU may be implemented within a network node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other network nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU, and RU also can be implemented as virtual units, such as a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU), among other examples.

Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized

in an IAB network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)) to facilitate scaling of communication systems by separating base station functionality into one or more units that can be individually deployed. A disaggregated base station may include functionality implemented across two or more units at various physical locations, as well as functionality implemented for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station can be configured for wired or wireless communication with at least one other unit of the disaggregated base station.

FIG. 3 is a diagram illustrating an example disaggregated base station architecture 300, in accordance with the present disclosure. The disaggregated base station architecture 300 may include a CU 310 that can communicate directly with a core network 320 via a backhaul link, or indirectly with the core network 320 through one or more disaggregated control units (such as a Near-RT RIC 325 via an E2 link, or a Non-RT RIC 315 associated with a Service Management and Orchestration (SMO) Framework 305, or both). A CU 310 may communicate with one or more DUs 330 via respective midhaul links, such as through F1 interfaces. Each of the DUs 330 may communicate with one or more RUs 340 via respective fronthaul links. Each of the RUs 340 may communicate with one or more UEs 120 via respective radio frequency (RF) access links. In some implementations, a UE 120 may be simultaneously served by multiple RUs 340.

Each of the units, including the CUs 310, the DUs 330, the RUs 340, as well as the Near-RT RICs 325, the Non-RT RICs 315, and the SMO Framework 305, may include one or more interfaces or be coupled with one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to one or multiple communication interfaces of the respective unit, can be configured to communicate with one or more of the other units via the transmission medium. In some examples, each of the units can include a wired interface, configured to receive or transmit signals over a wired transmission medium to one or more of the other units, and a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

In some aspects, the CU 310 may host one or more higher layer control functions. Such control functions can include radio resource control (RRC) functions, packet data convergence protocol (PDCP) functions, or service data adaptation protocol (SDAP) functions, among other examples. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 310. The CU 310 may be configured to handle user plane functionality (for example, Central Unit-User Plane (CU-UP) functionality), control plane functionality (for example, Central Unit-Control Plane (CU-CP) functionality), or a combination thereof. In some implementations, the CU 310 can be logically split into one or more CU-UP units and one or more CU-CP units. A CU-UP unit can communicate bidirectionally with a CU-CP unit via an interface, such as the E1 interface when implemented in an

O-RAN configuration. The CU 310 can be implemented to communicate with a DU 330, as necessary, for network control and signaling.

Each DU 330 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 340. In some aspects, the DU 330 may host one or more of a radio link control (RLC) layer, a MAC layer, and one or more high physical (PHY) layers depending, at least in part, on a functional split, such as a functional split defined by the 3GPP. In some aspects, the one or more high PHY layers may be implemented by one or more modules for forward error correction (FEC) encoding and decoding, scrambling, and modulation and demodulation, among other examples. In some aspects, the DU 330 may further host one or more low PHY layers, such as implemented by one or more modules for a fast Fourier transform (FFT), an inverse FFT (iFFT), digital beamforming, or physical random access channel (PRACH) extraction and filtering, among other examples. Each layer (which also may be referred to as a module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 330, or with the control functions hosted by the CU 310.

Each RU 340 may implement lower-layer functionality. In some deployments, an RU 340, controlled by a DU 330, may correspond to a logical node that hosts RF processing functions or low-PHY layer functions, such as performing an FFT, performing an iFFT, digital beamforming, or PRACH extraction and filtering, among other examples, based on a functional split (for example, a functional split defined by the 3GPP), such as a lower layer functional split. In such an architecture, each RU 340 can be operated to handle over the air (OTA) communication with one or more UEs 120. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) 340 can be controlled by the corresponding DU 330. In some scenarios, this configuration can enable each DU 330 and the CU 310 to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

The SMO Framework 305 may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework 305 may be configured to support the deployment of dedicated physical resources for RAN coverage requirements, which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework 305 may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) platform 390) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs 310, DUs 330, RUs 340, non-RT RICs 315, and Near-RT RICs 325. In some implementations, the SMO Framework 305 can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) 311, via an O1 interface. Additionally, in some implementations, the SMO Framework 305 can communicate directly with each of one or more RUs 340 via a respective O1 interface. The SMO Framework 305 also may include a Non-RT RIC 315 configured to support functionality of the SMO Framework 305.

The Non-RT RIC 315 may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including

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model training and updates, or policy-based guidance of applications/features in the Near-RT RIC 325. The Non-RT RIC 315 may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC 325. The Near-RT RIC 325 may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs 310, one or more DUs 330, or both, as well as an O-eNB, with the Near-RT RIC 325.

In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC 325, the Non-RT RIC 315 may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC 325 and may be received at the SMO Framework 305 or the Non-RT RIC 315 from non-network data sources or from network functions. In some examples, the Non-RT RIC 315 or the Near-RT RIC 325 may be configured to tune RAN behavior or performance. For example, the Non-RT RIC 315 may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework 305 (such as reconfiguration via an O1 interface) or via creation of RAN management policies (such as A1 interface policies).

As indicated above, FIG. 3 is provided as an example. Other examples may differ from what is described with regard to FIG. 3.

FIGS. 4A-4C are diagrams illustrating examples 400, 410, 420 of full duplex (FD) communication in accordance with the present disclosure. The example 400 of FIG. 4A includes a UE1 402 and two base stations (e.g., TRPs) 404-1, 404-2, where the UE1 402 is sending UL transmissions to base station 404-1 and is receiving DL transmissions from base station 404-2. In the example 400 of FIG. 4A, FD is enabled for the UE1 402, but not for the base stations 404-1, 404-2. The example 410 of FIG. 4B includes two UEs, shown as UE1 402-1 and UE2 402-2, and a base station 404, where the UE1 402-1 is receiving a DL transmission from the base station 404 and the UE2 402-2 is transmitting an UL transmission to the base station 404. In the example 410 of FIG. 4B, FD is enabled for the base station 404, but not for UE1 402-1 and UE2 402-2. The example 420 of FIG. 4C includes a UE1 402 and a base station 404, where the UE1 402 is receiving a DL transmission from the base station 404 and the UE1 402 is transmitting an UL transmission to the base station 404. In the example 420 of FIG. 4C, FD is enabled for both the UE1 402 and the base station 404.

As indicated above, FIGS. 4A-4C are provided as one or more examples. Other examples may differ from what is described with regard to FIGS. 4A-4C.

FIGS. 5A and 5B are diagrams illustrating a first example 500 and a second example 502 of full duplex communications based at least in part on multiple antenna panels, in accordance with the present disclosure.

A device performing full duplex transmissions may concurrently (and simultaneously) transmit and receive using a same frequency. Transmitting a first signal based at least in part on using a same frequency as a second signal being received may result in self-interference at a receiver of the device. “Self-interference” may refer to a transmission by a device that leaks into reception by the device. For example, “clutter” denotes a portion of a transmission signal that reflects off of, diffracts around, and/or scatters based at least in part on an object. Clutter may propagate toward a receiver and result in a degraded signal quality (e.g., of the intended

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second signal) at the receiver, such as a lower signal-to-interference-plus-noise ratio (SINR). An amount of clutter received by the receiver may be time-varying based on changes in a transmission channel, changes in an operating location, and/or atmospheric changes.

In some aspects, a device, such as a UE, may use multiple antenna panels to improve spherical transmission and/or reception coverage. Alternatively or additionally, the device may use multiple antenna panels to mitigate signal blockages, such as by switching transmission or reception at a blocked antenna panel to an unblocked antenna panel. An “antenna panel” may denote an antenna module that includes multiple antenna elements and may be implemented using any combination of hardware, software, and/or firmware. The multiple antenna components may be configured to collectively transmit one or more beams in a variety of directions as part of a phased array and, in some aspects, multiple antenna panels may be co-phased with one another to collectively form a phased array. Full duplex communications that are based at least in part on the use of multiple antenna panels may experience different self-interference.

The first example 500 includes a network node 110a and a UE 120a that communicate with one another based at least in part on full duplex transmissions (shown as rank=2 based at least in part on a full duplex transmission that includes a single downlink signal and a single uplink signal). As shown by the example 500, the UE 120a includes a first antenna panel 504a positioned at a first outer edge of the UE 120a, a second antenna panel 506a positioned at a second outer edge of the UE 120a, and a third antenna panel 508a positioned at a third outer edge of the UE 120a. In some aspects, the UE 120a may perform full duplex communications based at least in part on configuring the first antenna panel 504a to receive a downlink signal 510a from the network node 110a and the second antenna panel 506a to transmit an uplink signal 512a to the network node 110a. Based at least in part on the UE 120a and/or the network node 110a operating in a full duplex mode, the downlink signal and the uplink signal may use a same frequency.

In some aspects, the transmissions may reflect off one or more objects. To illustrate, the downlink signal 510a may reflect off a first object 514a (shown as a vehicle) and the uplink signal 512a may reflect off a second object 516a (shown as a structure). The reflections may result in self-interference at the UE 120a and/or the network node 110a. As one example, the uplink signal 512a may reflect off the second object 516a and generate clutter 518a.

The second example 502 includes a network node 110b and a UE 120b that communicate with one another based at least in part on full duplex transmissions (shown as rank=2 based at least in part on a full duplex transmission that includes a single downlink signal and a single uplink signal). The UE 120b includes multiple antenna panels, shown as a first antenna panel 504b positioned at a first outer edge of the UE 120b, a second antenna panel 506b positioned at a second outer edge of the UE 120b, and a third antenna panel 508b positioned at a third outer edge of the UE 120b. In the example 502, the UE 120b configures a first subarray of antenna elements of the first antenna panel 504b to receive a downlink signal 510b from the network node 110b and a second subarray of antenna elements of the first antenna panel 504b to transmit an uplink signal 512b to the network node 110b. The downlink signal 510b may reflect off a first object 514b (shown as a vehicle) and the uplink signal 512b may reflect off a second object 516b (shown as a structure) that generates clutter 518b.

A first amount of self-interference observed by the UE **120a** may differ from a second amount of self-interference observed by the UE **120b** based at least in part on the different antenna panel configurations used for full duplex communications. To illustrate, using separate antenna panels as shown by the example **500** may provide more isolation from self-interference (e.g., the clutter **518a**) relative to using subarrays of a same antenna panel as shown by the example **502** based at least in part on a physical separation between the two antenna panels. Accordingly, an antenna panel configuration used by a device for full duplex communications (e.g., a dedicated antenna panel for transmission, a dedicated antenna panel for reception, and/or an antenna panel partitioned to perform transmission and reception), as well as changing operating conditions as described above, may change an amount of self-interference observed by the device. Without information that indicates the antenna panel configuration and/or how the device observes self-interference, a configuring device (e.g., the network node **110**) may select a first full duplex configuration (e.g., an antenna set and beam selection) that results in reduced performance relative to a second full duplex configuration, such as reduced data throughput, increased data transfer latencies, and/or increased data recovery errors.

Some techniques and apparatuses described herein provide self-interference feedback for multiple antenna panels. A UE may receive a beam based at least in part on using at least a first antenna panel. Based at least in part on receiving the beam, the UE may transmit a rank indicator (RI) and UE assistance information (UAI). The UAI may include a self-interference metric associated with operating in a full duplex mode and/or based at least in part on using the first antenna panel for full duplex communication. In some aspects, the UE may generate the UAI based at least in part on receiving full duplex configuration information and/or self-interference configuration information.

In some aspects, a network node may receive an RI and/or UAI associated with a full duplex communication with a UE. As one example, the UAI may include a self-interference metric that is associated with the UE operating in a full duplex mode and/or based at least in part on the UE using at least a first antenna panel for the full duplex communication. The network node may transmit full duplex configuration information that is based at least in part on the RI and the UAI.

By indicating UAI with an RI, a UE may provide a network node with information about how the UE receives self-interference and/or how self-interference affects full duplex communication by the UE. Indicating UAI with the rank indicator also enables the UE to update the UAI as operating conditions change. A network node may subsequently select and/or update a full duplex configuration that improves signal quality at the UE and/or the network node based at least in part on the updated UAI. The improved signal quality may increase data throughput, reduce data transfer latencies, and/or reduce data recovery errors relative to a full duplex configuration that is not based on UAI and results in lower signal quality.

As indicated above, FIGS. **5A** and **5B** are provided as examples. Other examples may differ from what is described with regard to FIGS. **5A** and **5B**.

FIG. **6** is a diagram illustrating an example **600** of a wireless communication process between a network node (e.g., the network node **110**) and a UE (e.g., the UE **120**), in accordance with the present disclosure.

As shown by reference number **610**, a UE **120** may transmit, and a network node **110** may receive, capability

information. As one example, the UE **120** may transmit capability information that indicates that the UE supports full duplex communications. Alternatively or additionally, the UE may indicate, in the capability information, a number of antenna panels included in the UE **120**. In some aspects, the capability information may indicate antenna panel configuration information, such as a first antenna panel configuration that indicates that a first antenna panel is transmit only, a second antenna panel configuration that indicates that a second antenna panel is receive only, and/or a third antenna panel configuration that indicates that a third antenna panel is partitionable into subarrays. In some aspects, the antenna panel configuration information may indicate that a first subarray of the third antenna panel has been configured for transmission and a second subarray of the third antenna panel has been configured for reception. However, in other examples, the UE **120** may refrain from indicating a number of antenna panels and/or antenna panel configuration information in the capability information.

As shown by reference number **620**, the network node **110** may select a full duplex configuration for performing full duplex communications with the UE **120**. To illustrate, the network node **110** may select a beam for a full duplex communication based at least in part on the UE indicating support for full duplex communication (e.g., via capability information). Alternatively or additionally, the network node **110** may select one or more UE antenna panels as part of the full duplex configuration. For example, the network node **110** may select one or more directional antenna panels (e.g., a first antenna panel for transmission and a second antenna panel for reception) and/or a bi-directional antenna panel that is partitionable as further described. In some aspects, the network node **110** may select a preferred set of transmission configuration information (TCI) states as part of the full duplex communication configuration.

The network node **110** may transmit, as at least part of the full duplex configuration information, self-interference configuration information that indicates a configuration for measuring self-interference. As one example, the network node **110** may indicate to enable and/or disable generating a self-interference metric. As another example, the network node **110** may indicate, in the self-interference configuration information, any combination of a beam selection for generating the self-interference metric, an antenna panel selection for generating the self-interference metric, a beam presence to use for generating the self-interference, and/or a beam absence to use for generating the self-interference metric. The network node **110** may indicate a type of self-interference metric to generate, such as SINR and/or a rate reduction metric (e.g., a reduction factor and/or percentage based at least in part on a current data rate). In some aspects, the network node **110** may transmit the self-interference configuration information in a separate transmission than is used for indicating the full duplex configuration information.

As shown by reference number **630**, the network node **110** may transmit, and the UE **120** may receive, full duplex configuration information. The network node **110** may transmit the full duplex configuration (e.g., that may include or exclude the self-interference configuration information) using any combination of layer **1** signaling, a radio resource control (RRC) message, a medium access control (MAC) control element (CE), and/or downlink control information (DCI). For instance, the network node **110** may transmit multiple beam configurations and/or antenna configurations based at least in part on using an RRC message and subsequently activate a particular beam configuration and/or

antenna configuration based at least in part on DCI. Thus, while the example 600 shows the transmission of the full duplex configuration information as a single transaction, other aspects may include multiple transmissions.

As shown by reference number 640, the network node 110 and the UE 120 may transmit and/or receive one or more full duplex transmissions. In some aspects, the one or more full duplex transmissions may be based at least in part on the full duplex configuration information. For instance, the network node 110 may transmit a first beam (e.g., indicated by the full duplex configuration information) that the UE 120 may receive based at least in part on using at least a first antenna panel (e.g., indicated by the full duplex configuration information). In some aspects, the UE 120 may configure one or more antenna panels based at least in part on the full duplex configuration information, such as by configuring an antenna panel as a directional antenna panel or configuring the antenna panel into one or more subarrays of antennas as indicated by the full duplex configuration information. Alternatively or additionally, the network node 110 and/or the UE 120 may transmit a respective reference signal based at least in part on the full duplex configuration information (e.g., for measuring self-interference). The reference signal may include one or more synchronization signal block (SSB) signals, one or more channel state information reference signals (CSI-RSs), and/or one or more sounding reference signals (SRSs). Alternatively or additionally, the reference signals may include other types of configured reference signals.

As shown by reference number 650, the UE 120 may generate one or more self-interference metrics. As one example, the UE may generate an SINR metric based at least in part on receiving the beam transmitted by the network node 110 and/or based at least in part on using an antenna panel as indicated by the self-interference configuration information. Alternatively or additionally, the UE may generate a rate reduction metric based at least in part on receiving the beam as indicated by the full duplex configuration information and/or the self-interference configuration information.

In some aspects, a rank determined by the UE 120 may be associated with a first beam, and a self-interference metric generated by the UE 120 may be based at least in part on a second beam. For example, the UE 120 may determine a rank associated with receiving a first beam based at least in part on a first antenna panel, and generate a self-interference metric based at least in part on receiving a second beam using the first antenna panel. Alternatively or additionally, the UE 120 may generate multiple self-interference metrics, such as a first self-interference metric that is based at least in part on receiving the first beam using the first antenna panel, and a second self-interference metric that is based at least in part on receiving the second beam using the second antenna panel. In some aspects, the UE 120 may generate a self-interference metric based at least in part on a presence or an absence of a beam and/or signal. To illustrate, the UE 120 may generate a first metric that is based at least in part on a presence of a beam (e.g., the first beam and/or the second beam), and/or a second metric that is based at least in part on an absence of the beam.

As shown by reference number 660, the UE 120 may transmit, and the network node 110 may receive, UAI and an RI. For example, the UE 120 may transmit UAI that includes one or more self-interference metrics as further described with regard to the reference number 650. In some aspects,

the UAI and/or the RI is based at least in part on the one or more full duplex transmissions as shown with regard to reference number 640.

As shown by reference number 670, the network node 110 may select and/or update a full duplex configuration based at least in part on the UAI and/or RI received as described with regard to the reference number 660. In some aspects, the network node 110 may iteratively select and/or update the full duplex configuration based at least in part on iteratively receiving UAI with an RI from the UE 120. Iteratively receiving the UAI with the RI enables the network node 110 to update the full duplex configuration based at least in part on changing self-interference, and to improve signal quality associated with full duplex transmissions. Improved signal quality may increase data throughput, reduce data transfer latencies, and/or reduce data recovery errors (e.g., relative to a full duplex configuration that is not based on UAI).

As indicated above, FIG. 6 is provided as an example. Other examples may differ from what is described with regard to FIG. 6.

FIG. 7 is a diagram illustrating an example process 700 performed, for example, by a UE, in accordance with the present disclosure. Example process 700 is an example where the UE (e.g., UE 120) performs operations associated with self-interference feedback for multiple antenna panels.

As shown in FIG. 7, in some aspects, process 700 may include receiving a beam based at least in part on using at least a first antenna panel (block 710). For example, the UE (e.g., using communication manager 140 and/or reception component 902, depicted in FIG. 9) may receive a beam based at least in part on using at least a first antenna panel, as described above.

As further shown in FIG. 7, in some aspects, process 700 may include transmitting an RI and UAI. The UAI may include a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel (block 720). For example, the UE (e.g., using communication manager 140 and/or transmission component 904, depicted in FIG. 9) may transmit an RI and UAI. The UAI may include a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel, as described above.

Process 700 may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

In a first aspect, receiving the beam based at least in part on using the at least first antenna panel may include using a subarray of antenna elements included in the at least first antenna panel.

In a second aspect, alone or in combination with the first aspect, the RI is associated with the beam, and the self-interference metric is associated with receiving the beam based at least in part on using the at least first antenna panel.

In a third aspect, alone or in combination with one or more of the first and second aspects, the beam is a first beam, wherein the RI is associated with the first beam, and wherein the self-interference metric is associated with receiving a second beam based at least in part on using the at least first antenna panel.

In a fourth aspect, alone or in combination with one or more of the first through third aspects, the beam is a first beam, wherein the RI is associated with the first beam, wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and wherein the

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self-interference metric is associated with receiving a second beam based at least in part on using the second antenna panel.

In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, process **700** includes receiving self-interference configuration information, and generating the self-interference metric based at least in part on the self-interference configuration information.

In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, the self-interference configuration information indicates at least one of an antenna panel selection, a beam selection.

In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, the at least first antenna panel includes the first antenna panel and a second antenna panel, and the antenna panel selection indicates the first antenna panel or the second antenna panel.

In an eighth aspect, alone or in combination with one or more of the first through seventh aspects, the beam received based at least in part on using the at least first antenna panel is a first beam, and the beam selection indicates the first beam or a second beam.

In a ninth aspect, alone or in combination with one or more of the first through eighth aspects, the self-interference metric includes at least one of a signal-to-interference-plus-noise ratio, or a rate reduction.

In a tenth aspect, alone or in combination with one or more of the first through ninth aspects, the self-interference metric includes at least one of a first metric that is based at least in part on a presence of the beam, or a second metric that is based at least in part on an absence of the beam.

In an eleventh aspect, alone or in combination with one or more of the first through tenth aspects, the RI is associated with the beam.

In a twelfth aspect, alone or in combination with one or more of the first through eleventh aspects, process **700** includes transmitting, prior to receiving the beam, capability information that indicates a number of antenna panels supported by the UE.

Although FIG. 7 shows example blocks of process **700**, in some aspects, process **700** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 7. Additionally, or alternatively, two or more of the blocks of process **700** may be performed in parallel.

FIG. 8 is a diagram illustrating an example process **800** performed, for example, by a network node, in accordance with the present disclosure. Example process **800** is an example where the network node (e.g., network node **110**) performs operations associated with self-interference feedback for multiple antenna panels.

As shown in FIG. 8, in some aspects, process **800** may include receiving an RI and UAI associated with a UE. The UAI may include a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel (block **810**). For example, the network node (e.g., using communication manager **150** and/or reception component **1002**, depicted in FIG. 10) may receive an RI and UAI associated with UE. The UAI may include a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel, as described above.

As further shown in FIG. 8, in some aspects, process **800** may include transmitting full duplex configuration information that is based at least in part on the RI and the UAI (block **820**). For example, the network node (e.g., using commu-

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nication manager **150** and/or transmission component **1004**, depicted in FIG. 10) may transmit full duplex configuration information that is based at least in part on the RI and the UAI, as described above.

Process **800** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

In a first aspect, process **800** includes transmitting a beam based at least in part on the full-duplex configuration.

In a second aspect, alone or in combination with the first aspect, the RI is associated with a beam, and the self-interference metric is associated with the UE receiving the beam based at least in part on using the at least first antenna panel.

In a third aspect, alone or in combination with one or more of the first and second aspects, the RI is associated with a first beam, and the self-interference metric is associated with the UE receiving a second beam based at least in part on using the at least first antenna panel.

In a fourth aspect, alone or in combination with one or more of the first through third aspects, the RI is associated with a first beam, wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and wherein the self-interference metric is associated with the UE receiving a second beam based at least in part on using the second antenna panel.

In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, process **800** includes transmitting self-interference configuration information, wherein the self-interference metric is based at least in part on the self-interference configuration information.

In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, the self-interference configuration information indicates at least one of an antenna panel selection, a beam selection.

In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, process **800** includes receiving capability information that indicates a number of antenna panels supported by the UE, wherein the self-interference configuration information is based at least in part on the capability information.

In an eighth aspect, alone or in combination with one or more of the first through seventh aspects, the self-interference metric includes at least one of a signal-to-interference-plus-noise ratio, or a rate reduction.

In a ninth aspect, alone or in combination with one or more of the first through eighth aspects, the self-interference metric includes at least one of a first metric that is based at least in part on a presence of a beam, or a second metric that is based at least in part on an absence of the beam.

Although FIG. 8 shows example blocks of process **800**, in some aspects, process **800** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 8. Additionally, or alternatively, two or more of the blocks of process **800** may be performed in parallel.

FIG. 9 is a diagram of an example apparatus **900** for wireless communication, in accordance with the present disclosure. The apparatus **900** may be a UE, or a UE may include the apparatus **900**. In some aspects, the apparatus **900** includes a reception component **902** and a transmission component **904**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **900** may communicate with another apparatus **906** (such as a UE, a base station, or another wireless communication device)

using the reception component **902** and the transmission component **904**. As further shown, the apparatus **900** may include the communication manager **140**. The communication manager **140** may include one or more of a self-interference manager component **908**, among other examples.

In some aspects, the apparatus **900** may be configured to perform one or more operations described herein in connection with FIGS. 5-8. Additionally, or alternatively, the apparatus **900** may be configured to perform one or more processes described herein, such as process **700** of FIG. 7, or a combination thereof. In some aspects, the apparatus **900** and/or one or more components shown in FIG. 9 may include one or more components of the UE described in connection with FIG. 2. Additionally, or alternatively, one or more components shown in FIG. 9 may be implemented within one or more components described in connection with FIG. 2. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

The reception component **902** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **906**. The reception component **902** may provide received communications to one or more other components of the apparatus **900**. In some aspects, the reception component **902** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **900**. In some aspects, the reception component **902** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2.

The transmission component **904** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **906**. In some aspects, one or more other components of the apparatus **900** may generate communications and may provide the generated communications to the transmission component **904** for transmission to the apparatus **906**. In some aspects, the transmission component **904** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **906**. In some aspects, the transmission component **904** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. In some aspects, the transmission component **904** may be co-located with the reception component **902** in a transceiver.

The reception component **902** may receive a beam based at least in part on using at least a first antenna panel. The transmission component **904** may transmit an RI and UAI. The UAI may include a self-interference metric associated

with operating in a full duplex mode based at least in part on using the at least first antenna panel.

The self-interference manager component **908**, by way of the reception component **902**, may receive self-interference configuration information. In some aspects, the self-interference manager component **908** may generate the self-interference metric based at least in part on the self-interference configuration information.

The transmission component **904** may transmit, prior to receiving the beam, capability information that indicates a number of antenna panels supported by the UE.

The number and arrangement of components shown in FIG. 9 are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. 9. Furthermore, two or more components shown in FIG. 9 may be implemented within a single component, or a single component shown in FIG. 9 may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. 9 may perform one or more functions described as being performed by another set of components shown in FIG. 9.

FIG. 10 is a diagram of an example apparatus **1000** for wireless communication, in accordance with the present disclosure. The apparatus **1000** may be a network node, or a network node may include the apparatus **1000**. In some aspects, the apparatus **1000** includes a reception component **1002** and a transmission component **1004**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **1000** may communicate with another apparatus **1006** (such as a UE, a base station, or another wireless communication device) using the reception component **1002** and the transmission component **1004**. As further shown, the apparatus **1000** may include the communication manager **150**. The communication manager **150** may include one or more of a self-interference manager component **1008**, among other examples.

In some aspects, the apparatus **1000** may be configured to perform one or more operations described herein in connection with FIGS. 5-8. Additionally, or alternatively, the apparatus **1000** may be configured to perform one or more processes described herein, such as process **800** of FIG. 8, or a combination thereof. In some aspects, the apparatus **1000** and/or one or more components shown in FIG. 10 may include one or more components of the network node described in connection with FIG. 2. Additionally, or alternatively, one or more components shown in FIG. 10 may be implemented within one or more components described in connection with FIG. 2. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

The reception component **1002** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **1006**. The reception component **1002** may provide received communications to one or more other components of the apparatus **1000**. In some aspects, the reception component **1002** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing,

deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **1000**. In some aspects, the reception component **1002** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the network node described in connection with FIG. 2.

The transmission component **1004** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **1006**. In some aspects, one or more other components of the apparatus **1000** may generate communications and may provide the generated communications to the transmission component **1004** for transmission to the apparatus **1006**. In some aspects, the transmission component **1004** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **1006**. In some aspects, the transmission component **1004** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the network node described in connection with FIG. 2. In some aspects, the transmission component **1004** may be co-located with the reception component **1002** in a transceiver.

The self-interference manager component **1008** may receive, by way of the reception component **1002**, an RI and UAI associated with a UE. The UAI may include a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel. The self-interference manager component **1008** may select and transmit, by way of the transmission component **1004**, full-duplex configuration information that is based at least in part on the RI and the UAI.

The transmission component **1004** may transmit a beam based at least in part on the full duplex configuration information.

The self-interference manager component **1008** may transmit, by way of the transmission component **1004**, self-interference configuration information, wherein the self-interference metric is based at least in part on the self-interference configuration information.

The self-interference manager component **1008** may receive, by way of the reception component **1002**, capability information that indicates a number of antenna panels supported by the UE, wherein the self-interference configuration information is based at least in part on the capability information.

The number and arrangement of components shown in FIG. 10 are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. 10. Furthermore, two or more components shown in FIG. 10 may be implemented within a single component, or a single component shown in FIG. 10 may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. 10 may perform one or more functions described as being performed by another set of components shown in FIG. 10.

The following provides an overview of some Aspects of the present disclosure:

Aspect 1: A method of wireless communication performed by a user equipment (UE), comprising: receiving a beam based at least in part on using at least a first antenna panel; and transmitting a rank indicator and user equipment assistance information (UAI), the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel.

Aspect 2: The method of Aspect 1, wherein the receiving the beam based at least in part on using the at least first antenna panel comprises: using a subarray of antenna elements included in the at least first antenna panel.

Aspect 3: The method of Aspect 1 or Aspect 2, wherein the rank indicator is associated with the beam, and wherein the self-interference metric is associated with receiving the beam based at least in part on using the at least first antenna panel.

Aspect 4: The method of any one of Aspects 1-3, wherein the beam is a first beam, wherein the rank indicator is associated with the first beam, and wherein the self-interference metric is associated with receiving a second beam based at least in part on using the at least first antenna panel.

Aspect 5: The method of any one of Aspects 1-3, wherein the beam is a first beam, wherein the rank indicator is associated with the first beam, wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and wherein the self-interference metric is associated with receiving a second beam based at least in part on using the second antenna panel.

Aspect 6: The method of any one of Aspects 1-5, further comprising: receiving self-interference configuration information; and generating the self-interference metric based at least in part on the self-interference configuration information.

Aspect 7: The method of Aspect 6, wherein the self-interference configuration information indicates at least one of: an antenna panel selection, a beam selection.

Aspect 8: The method of Aspect 7, wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and wherein the antenna panel selection indicates the first antenna panel or the second antenna panel.

Aspect 9: The method of Aspect 7 or Aspect 8, wherein the beam received based at least in part on using the at least first antenna panel is a first beam, and wherein the beam selection indicates the first beam or a second beam.

Aspect 10: The method of any one of Aspects 1-9, wherein the self-interference metric comprises at least one of: a signal-to-interference-plus-noise ratio, or a rate reduction.

Aspect 11: The method of any one of Aspects 1-10, wherein the self-interference metric comprises at least one of: a first metric that is based at least in part on a presence of the beam, or a second metric that is based at least in part on an absence of the beam.

Aspect 12: The method of Aspect 11, wherein the rank indicator is associated with the beam.

Aspect 13: The method of any one of Aspects 1-12, further comprising: transmitting, prior to receiving the beam, capability information that indicates a number of antenna panels supported by the UE.

Aspect 14: A method of wireless communication performed by a network node, comprising: receiving a rank indicator and user equipment assistance information (UAI) associated with a user equipment (UE), the UAI comprising

a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel; and transmitting full duplex configuration information that is based at least in part on the rank indicator and the UAI.

Aspect 15: The method of Aspect 14, further comprising: transmitting a beam based at least in part on the full duplex configuration information.

Aspect 16: The method of Aspect 14 or Aspect 15, wherein the rank indicator is associated with a beam, and wherein the self-interference metric is associated with the UE receiving the beam based at least in part on using the at least first antenna panel.

Aspect 17: The method of any one of Aspects 14-16, wherein the rank indicator is associated with a first beam, and wherein the self-interference metric is associated with the UE receiving a second beam based at least in part on using the at least first antenna panel.

Aspect 18: The method of any one of Aspects 14-16, wherein the rank indicator is associated with a first beam, wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and wherein the self-interference metric is associated with the UE receiving a second beam based at least in part on using the second antenna panel.

Aspect 19: The method of any one of Aspects 14-18, further comprising: transmitting self-interference configuration information, wherein the self-interference metric is based at least in part on the self-interference configuration information.

Aspect 20: The method of Aspect 19, wherein the self-interference configuration information indicates at least one of: an antenna panel selection, a beam selection.

Aspect 21: The method of Aspect 19 or Aspect 20, further comprising: receiving capability information that indicates a number of antenna panels supported by the UE, wherein the self-interference configuration information is based at least in part on the capability information.

Aspect 22: The method of any one of Aspects 14-21, wherein the self-interference metric comprises at least one of: a signal-to-interference-plus-noise ratio, or a rate reduction.

Aspect 23: The method of any one of Aspects 14-22, wherein the self-interference metric comprises at least one of: a first metric that is based at least in part on a presence of a beam, or a second metric that is based at least in part on an absence of the beam.

Aspect 24: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 1-13.

Aspect 25: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 14-23.

Aspect 26: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 1-13.

Aspect 27: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 14-23.

Aspect 28: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 1-13.

Aspect 29: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 14-23.

Aspect 30: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 1-13.

Aspect 31: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 14-23.

Aspect 32: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 1-13.

Aspect 33: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 14-23.

The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects.

As used herein, the term "component" is intended to be broadly construed as hardware and/or a combination of hardware and software. "Software" shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, and/or functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. As used herein, a "processor" is implemented in hardware and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the aspects. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code, since those skilled in the art will understand that software and hardware can be designed to implement the systems and/or methods based, at least in part, on the description herein.

As used herein, "satisfying a threshold" may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various aspects. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. The disclosure of various aspects includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase

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referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the terms “set” and “group” are intended to include one or more items and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

What is claimed is:

1. An apparatus for wireless communication at a user equipment (UE), the apparatus comprising:
 - a memory; and
 - one or more processors, coupled to the memory and configured to cause the UE to:
 - receive a beam based at least in part on using at least a first antenna panel; and
 - transmit a rank indicator and user equipment assistance information (UAI), the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel, wherein the self-interference metric comprises a rate reduction percentage, the rate reduction percentage based at least in part on a current data rate.
2. The apparatus of claim 1, wherein the one or more processors are further configured to cause the UE to:
 - use a subarray of antennas included in the at least first antenna panel.
3. The apparatus of claim 1, wherein the rank indicator is associated with the beam, and
 - wherein the self-interference metric is associated with receiving the beam based at least in part on using the at least first antenna panel.
4. The apparatus of claim 1, wherein the beam is a first beam,
 - wherein the rank indicator is associated with the first beam, and
 - wherein the self-interference metric is associated with receiving a second beam based at least in part on using the at least first antenna panel.
5. The apparatus of claim 1, wherein the beam is a first beam,
 - wherein the rank indicator is associated with the first beam,
 - wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and

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wherein the self-interference metric is associated with receiving a second beam based at least in part on using the second antenna panel.

6. The apparatus of claim 1, wherein the one or more processors are further configured to cause the UE to:
 - receive self-interference configuration information; and
 - generate the self-interference metric based at least in part on the self-interference configuration information.
7. The apparatus of claim 1, wherein the UAI comprises a signal-to-interference-plus-noise ratio.
8. The apparatus of claim 1, wherein the self-interference metric
 - is based at least in part on a presence of the beam or
 - on an absence of the beam.
9. The apparatus of claim 8, wherein the rank indicator is associated with the beam.
10. The apparatus of claim 1, wherein the one or more processors are further configured to cause the UE to:
 - transmit, prior to receiving the beam, capability information that indicates a number of antenna panels supported by the UE.
11. An apparatus for wireless communication at a network node, comprising:
 - a memory; and
 - one or more processors, coupled to the memory and configured to cause the network node to:
 - receive a rank indicator and user equipment assistance information (UAI) associated with a user equipment (UE), the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel, wherein the self-interference metric comprises a rate reduction percentage, the rate reduction percentage based at least in part on a current data rate; and
 - transmit a full-duplex configuration that is based at least in part on the rank indicator and the UAI.
12. The apparatus of claim 11, wherein the one or more processors are further configured to cause the network node to:
 - transmit a beam based at least in part on the full-duplex configuration.
13. The apparatus of claim 11, wherein the rank indicator
 - is associated with a beam, and
 - wherein the self-interference metric is associated with the UE receiving the beam based at least in part on using the at least first antenna panel.
14. The apparatus of claim 11, wherein the rank indicator
 - is associated with a first beam, and
 - wherein the self-interference metric is associated with the UE receiving a second beam based at least in part on using the at least first antenna panel.
15. The apparatus of claim 11, wherein the rank indicator
 - is associated with a first beam,
 - wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and
 - wherein the self-interference metric is associated with the UE receiving a second beam based at least in part on using the second antenna panel.
16. The apparatus of claim 11, wherein the one or more processors are further configured to cause the network node to:
 - transmit self-interference configuration information,
 - wherein the self-interference metric is based at least in part on the self-interference configuration information.

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17. The apparatus of claim 16, wherein the self-interference configuration information indicates at least one of:
an antenna panel selection, or
a beam selection.

18. The apparatus of claim 16, wherein the one or more processors are further configured to cause the network node to:

receive capability information that indicates a number of antenna panels supported by the UE,
wherein the self-interference configuration information is based at least in part on the capability information.

19. The apparatus of claim 11, wherein the self-interference metric

is based at least in part on a presence of a beam or on an absence of the beam.

20. A method of wireless communication performed by a user equipment (UE), comprising:

receiving a beam based at least in part on using at least a first antenna panel; and

transmitting a rank indicator and user equipment assistance information (UAI), the UAI comprising a self-interference metric associated with operating in a full duplex mode based at least in part on using the at least first antenna panel, wherein the self-interference metric comprises a rate reduction percentage, the rate reduction percentage based at least in part on a current data rate.

21. The method of claim 20, wherein the beam is a first beam,

wherein the rank indicator is associated with the first beam, and

wherein the self-interference metric is associated with receiving a second beam based at least in part on using the at least first antenna panel.

22. The method of claim 20, wherein the beam is a first beam,

wherein the rank indicator is associated with the first beam,

wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and

wherein the self-interference metric is associated with receiving a second beam based at least in part on using the second antenna panel.

23. The method of claim 20, further comprising:

receiving self-interference configuration information; and
generating the self-interference metric based at least in part on the self-interference configuration information.

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24. The method of claim 23, wherein the self-interference configuration information indicates at least one of:
an antenna panel selection, or
a beam selection.

25. The method of claim 24, wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and

wherein the antenna panel selection indicates the first antenna panel or the second antenna panel.

26. A method of wireless communication performed by a network node, comprising:

receiving a rank indicator and user equipment assistance information (UAI) associated with a user equipment (UE), the UAI comprising a self-interference metric associated with the UE operating in a full duplex mode based at least in part on using at least a first antenna panel, wherein the self-interference metric comprises a rate reduction percentage, the rate reduction percentage based at least in part on a current data rate; and
transmitting a full-duplex configuration that is based at least in part on the rank indicator and the UAI.

27. The method of claim 26, wherein the rank indicator is associated with a beam, and

wherein the self-interference metric is associated with the UE receiving the beam based at least in part on using the at least first antenna panel.

28. The method of claim 26, wherein the rank indicator is associated with a first beam, and

wherein the self-interference metric is associated with the UE receiving a second beam based at least in part on using the at least first antenna panel.

29. The method of claim 26, wherein the rank indicator is associated with a first beam,

wherein the at least first antenna panel includes the first antenna panel and a second antenna panel, and

wherein the self-interference metric is associated with the UE receiving a second beam based at least in part on using the second antenna panel.

30. The method of claim 26, further comprising:

receiving capability information that indicates a number of antenna panels supported by the UE; and
transmitting self-interference configuration information, wherein the self-interference configuration information is based at least in part on the capability information, and

wherein the self-interference metric is based at least in part on the self-interference configuration information.

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